

NZME: Driving audience engagement through powerful analytics and machine learning

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ABOUT NZME



NZME uses Google Analytics 360 and ML to run its Karri house built recommendation improve its ability audience engage earn ad impressions

About NZME

NZME combines radio, print, and digital across more than 50 of New Zealand's leading media brands to reach 3.6 million New Zealanders. Its flagship product is the New Zealand Herald with 160 years of newspaper publishing history that delivers trusted quality journalism and is a powerful presence in the digital era.

Industries: Media & Entertainment

convert readers into subscribers.

Location: New Zealand

Google Cloud results

- Data transformations that once took 2 hours took just 9 minutes
- Personalized recommendations delivered in 80 milliseconds
- 1 billion recommendations delivered with zero downtime
- 21% increase in recommendation click-through rate
- 9% increase in average session duration

Enhanced
recommendations
and

, Looker Studio

Google Cloud (<https://cloud.google.com/>)

Cloud Memorystore

Google Analytics

(<https://marketingplatform.google.com/about/resources/analytics-product-overview/>)

App Engine (<https://cloud.google.com/appengine/>)

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BigQuery (<https://cloud.google.com/bigquery/>)

AI & ML (<https://cloud.google.com/products/ai>)

Looker Studio

(<https://cloud.google.com/looker-studio>)

Cloud Memorystore

(<https://cloud.google.com/memorystore/>)

Exploring the latest updates on our favorite news outlets is a great way to keep up with what's happening in the world around us. For the teams behind the scenes at these publishers, ensuring they deliver the right articles at the right time is crucial to

support their mission to turn casual readers into loyal subscribers.

At NZME (<https://www.nzme.co.nz/>), New Zealand Media & Entertainment, this mission is about connecting a long history with a clear path into the future. Founded in 2014, the company is fairly young but has brought together existing magazines, radio stations, local newspapers, and the New Zealand Herald, a masthead with 160 years of history. Across its print, radio, and digital platforms, NZME reaches more than 70% of New Zealand's population each month across a wide range of demographics.

NZME has a key focus on making the NZ Herald a 'subscriber first' publication, with an emphasis on delivering a quality of experience that encourages readers to become paying supporters. To achieve this, NZME's digital team is working hard to elevate its data and analytics processes to better serve its audience.

"We've come to the point where digital advertising revenue has overtaken print advertising revenue, which is great in terms of sustainability," says Andy Wylie, Head of Data & Analytics at NZME. "Changes in the privacy landscape are a challenge for digital, but also an opportunity to build greater value from the first-party data that our readers share with us."

**Getting the fundamentals right
with reliable analytics
infrastructure**

NZME had been struggling to manage its analytics services. As the company developed its analytics platform, the demands on infrastructure management became a challenge for the IT department.

"There were challenges with maintaining the infrastructure and ensuring its ongoing performance," says Wylie. "With the operational overhead, difficult administration and a lack of scalability making it hard to add workloads, we moved to Google Cloud (<https://cloud.google.com/>)."

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The move took place in 2017, with NZME quickly finding the combination of Google Analytics 360 (<https://marketingplatform.google.com/about/analytics-360/>)

and its direct pipeline into BigQuery (<https://cloud.google.com/bigquery>) removed the management overhead from the technical teams while providing immediate improvements in utility.

"If you want to run a machine learning model on user data, you need access to all the granular data. It can't be highly aggregated," says Wylie. "Out of the box, Google handles the pipeline and streams it directly into BigQuery from GA 360. We're generating value right away."

"Our earlier recommendation engine was quite a compute intensive workload and took about two hours to complete within our old cluster. When we ported that code over to BigQuery, with just a few rewrites adding some efficiency, they were completed in nine minutes."

Previously, those hours in processing made it too hard to spend time iterating on their solutions. But with such rapid execution on BigQuery the team

could now review and revise results and make continuous improvements to their performance.

Predictive analytics elevate reader recommendations

For NZME, the flow of data between GA 360 into BigQuery is an especially powerful opportunity to power predictive analytics for their digital properties.

Using [BigQuery ML](#)

(<https://cloud.google.com/bigquery/docs#docs>), NZME was quickly getting predictive analytics from out-of-the-box algorithms.

"It only took a few lines of code," says Wylie. "You just shape up the data within the BigQuery table, run the algorithm, get the results, and start working with the data from there. The time-to-value was really impressive."

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Google offered assistance to NZME through its [Google News Initiative](https://newsinitiative.withgoogle.com/) (<https://newsinitiative.withgoogle.com/>), helping the team to identify use cases that held potential for driving new value for the business. The first project was to build an improved recommendation engine powered by BigQuery ML.

"Google has really looked after us," says Wylie. "They've made specialised experts available when we've hit barriers. They've given us advice on architecture. It saved us a lot of time so we avoided going down the wrong track."

After helping NZME build its first version, the publisher then continued into a second generation concept of its own design, called Karma. Across these two engines the company saw its recommendations deliver a 9% lift in average session duration and a 21% improvement in recommendation click throughs, generating millions of additional pageviews per month.

"The higher user engagement is, the more ad impressions are generated and the more likely they are to become a premium subscriber. So it's really important for our strategic goals," says Wylie.

Using granular user data, NZME can make recommendations for every individual user based on other users with similar reading history. This is applied alongside natural language processing to understand the context of the article they're currently reading, while further filtering is applied to remove duplicate articles the reader has previously viewed.

The data is run through BigQuery ML with results cached in [Memorystore for Redis](https://cloud.google.com/memorystore) (<https://cloud.google.com/memorystore>), with calls to [App Engine](https://cloud.google.com/appengine) (<https://cloud.google.com/appengine>) to retrieve the specific results for the user and deliver to the webpage they're viewing. Wylie says the round trip to deliver personalized recommendations to article pages is around 80 milliseconds, with over one billion recommendations delivered to date with zero downtime.

Converting data into powerful dashboards for business teams

Beyond its Karma recommendation engine, NZME needs its analytics to deliver insights across various internal teams to help sales, marketing, and other departments set targets and monitor progress toward goals. With BigQuery, it's become easier than in the past to make dashboards accessible to all.

"Democratizing access to data is a big part of the strategy for my team," says Wylie. "We can easily create custom metrics and model segments. We just click a button for any table in BigQuery and you visualize it in [Looker Studio](https://cloud.google.com/looker-studio) (<https://cloud.google.com/looker-studio>) and share it."

"Every analytics team has limited resources for responding to business analyst questions. Instead of just answering their queries, by providing a dashboard you give them the answer this time but also into the future."

The data team is also using the potential of [Dataform](https://cloud.google.com/dataform) (<https://cloud.google.com/dataform>) to better codify data modeling processes for analysts across NZME, allowing them to write queries more efficiently within a clear framework. The process will add version control and automated documentation to reveal data lineage for better clarity around how insights are being produced.

"You might have a single table output from a complex SQL query and data model, but it's pulling from eight source tables and three staging tables. This produces the diagram so you can drill in and understand the details," says Wylie. "It's going to improve our data quality and governance practices. It's a bit of a game changer."

The future of ML supported publishing engagement

NZME continues to look at how it can evolve its Karma recommendation engine. Wylie and his team

are exploring options to better understand how quickly different kinds of content "burnout," a traffic incident may hold little value within hours, while a lifestyle article remains valuable for weeks. The team is also testing time of day models, to see how the best recommendations might vary between business hours and evening reading sessions.

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"I think Google Cloud is the obvious choice in media and marketing," says Wylie. "You need to work with raw data if you want to get predictive analytics right and we've seen big gains through BigQuery ML. And with new features releasing all the time and support for more and more models, our hands-on experience has shown it can really deliver."